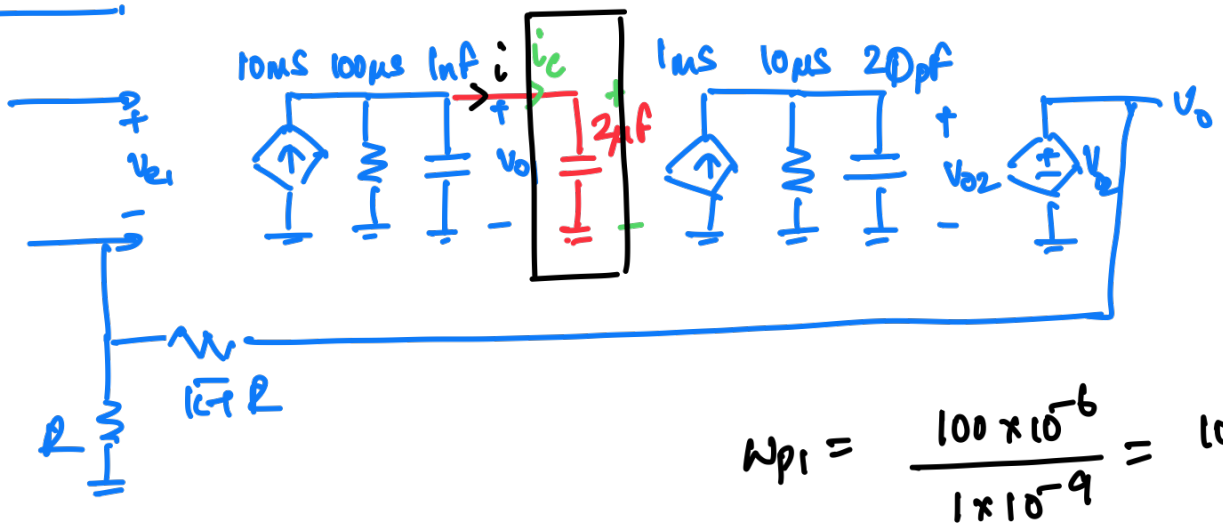
Lecture #16

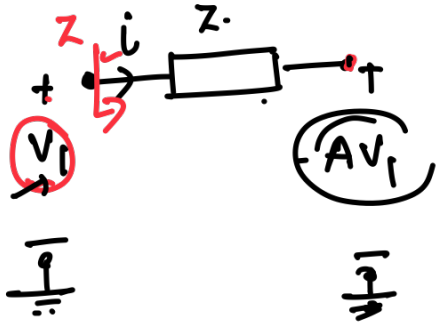


$$\omega_{p1} = \frac{100\times10^{-6}}{1\times10^{-9}} = 10^5 \text{ rad/s}$$

$$\omega_{p2} = \frac{10\times10^{-6}}{2p\times10^{-12}} = 5\times10^5 \text{ rad/s}$$

$i_c = V_{o1}\, sC_x$

$$\underline{V_c = \frac{i}{\quad}} = V_{o1}\, sC_x = \frac{V_{o1}}{Z_1} = \frac{\alpha V_{o1}}{\beta Z_1}$$



$$i = \frac{V_i - (-AV_i)}{Z_1} = \frac{V_i}{Z_1/(1+A)}$$

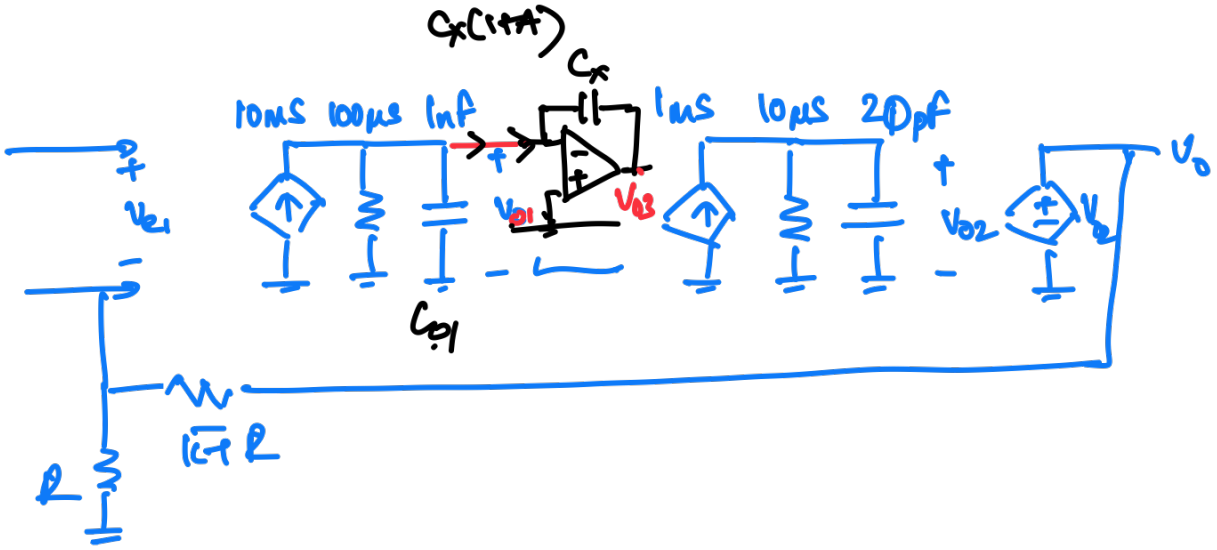
$$\frac{V_i}{i} = \frac{Z_1}{(1+A)}$$

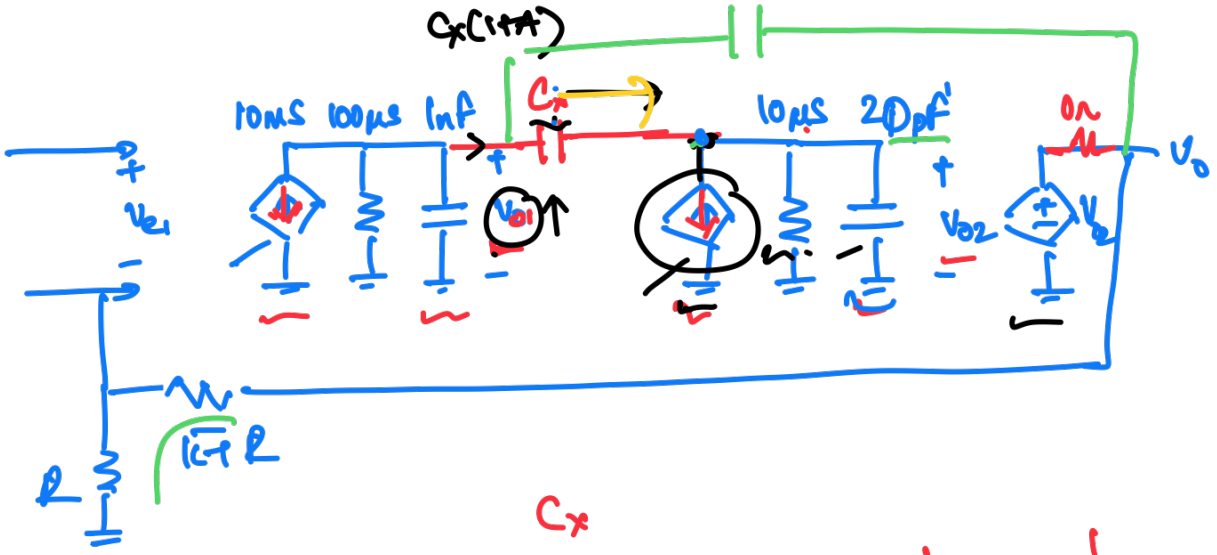
if $Z_1 = \frac{1}{sC_x}$

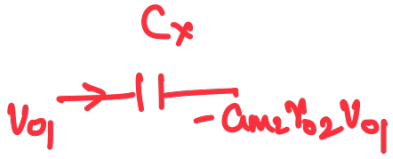
$$\frac{V_1}{i} = \frac{1}{s\,C_x(1+A)} = \frac{1}{sC_y}$$

$$C_y = C_x(1+A)$$

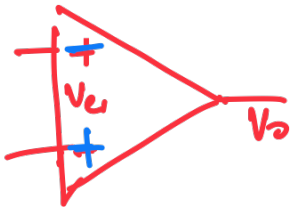
"Miller Multiplication"



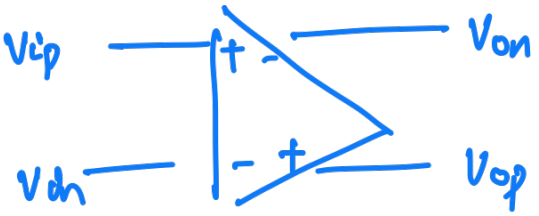




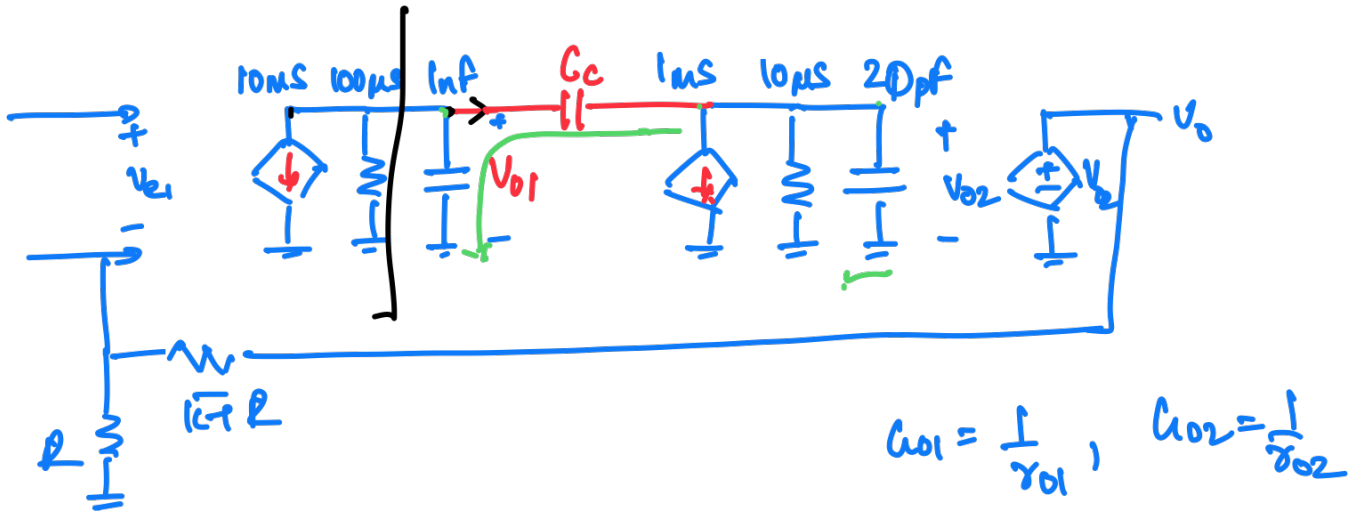
$$i = \frac{1}{sC_x(1+A_{v2}\gamma_{o2})} V_{o1}$$



$$V_o = AV_e$$



$$V_{op} - V_{on} = A(V_{ip} - V_n)$$



$$G_{m1}V_i + \frac{V_{01}}{r_{01}} + V_{01}sC_{01} + (V_{01} - V_{02})sC_c = 0$$

$$G_{m1}V_i + V_{01}(G_{o1} + sC_{01} + sC_c) - V_{02}sC_c = 0 \quad -(1)$$

$$G_{m2}V_{01} + V_{02}(G_{o2} + sC_{02}) + (V_{02} - V_{01})sC_c = 0$$

$$(G_{m2} - sC_c)V_{01} + V_{02}(G_{o2} + sC_{02} + sC_c) = 0 \quad -(2)$$

$$G_{m1}V_i - V_{02}sC_c + (G_{o1} + sC_{01} + sC_c) \times -\frac{V_{02}(G_{o2} + sC_{02} + sC_c)}{(G_{m2} - sC_c)}$$

$$+ V_{02}\left[\frac{sC_cG_{m2} - s^2C_c^2 + (G_{o1} + sC_{01} + sC_c)(G_{o2} + sC_{02} + sC_c)}{(G_{m2} - sC_c)}\right]$$

$$= G_{m1}V_i$$

$$\frac{V_{02}}{V_i} = \frac{G_{m1}(G_{m2} - sC_c)}{G_{o1}G_{o2} + s(C_{02} + C_c)G_{o1} + sC_{01}G_{o2} + s^2C_{01}C_c + sC_cG_{o2} + s^2C_{01}C_{02} + s^2C_cC_{02} + sC_cG_{m2}}$$

$$LG = \frac{1}{K} \frac{G_{m1}G_{m2}\left(1 - sC_c/G_{m2}\right)}{G_{o1}G_{o2} + s\left((G_{o2}+C_c)G_{o1} + (G_{o1}+C_c)G_{o2} + C_cG_{m2}\right) + s^2\left(C_{o1}C_{o2} + C_cC_{o2} + C_cC_{o1}\right)}$$

$$= \frac{1}{K} \frac{G_{m1}r_{o1}\,G_{m2}r_{o2}\left(1 - sC_c/G_{m2}\right)}{1 + s\left((C_{o2}+C_c)r_{o2} + (C_{o1}+C_c)r_{o1} + C_cG_{m2}r_{o1}r_{o2}\right) + s^2\left(C_{o1}C_{o2} + C_cC_{o2} + C_cC_{o1}\right)r_{o1}r_{o2}}$$

$$= \frac{1}{K} \frac{G_{m1}r_{o1}\,G_{m2}r_{o2}\left(1 - sC_c/G_{m2}\right)}{1 + s\left((C_{o2}+C_c)r_{o2} + \underbrace{(C_{o1}+C_c + C_cG_{m2}r_{o2})}_{C_c(1+G_{m2}r_{o2})}r_{o1}\right) + s^2\left(C_{o1}C_{o2} + C_cC_{o2} + C_cC_{o1}\right)r_{o1}r_{o2}}$$

$$\left\{C_{o1} + C_c\left(1 + G_{m2}r_{o2}\right)\right\}r_{o1}$$

$$D(s) = s^2\left(C_{o1}(C_{o2}+C_c) + C_cC_{o2}\right)r_{o1}r_{o2} + s\left((C_{o2}+C_c)r_{o2} + \left(C_{o1} + C_c(1+G_{m2}r_{o2})\right)r_{o1}\right) + 1 = ax^2 + bx + c$$

(higher freq. pole)

$$p_2' = -\frac{\left((C_{o2}+C_c)r_{o2} + \left(C_{o1} + C_c(1+G_{m2}r_{o2})\right)r_{o1}\right)}{\left(C_{o1}(C_{o2}+C_c) + C_cC_{o2}\right)r_{o1}r_{o2}}$$

$$p_1' = -\frac{1}{\left((C_{o2}+C_c)\,r_{o2} + \left(C_{o1} + C_c\left(1+g_{m2}r_{o2}\right)\right)r_{o1}\right)}$$

(Dominant pole)

$C_c\left(1+g_{m2}r_{o2}\right) \gg C_{o1}$

$$p_1 = -\frac{1}{r_{o1}C_{o1}} > p_1'$$

$$p_2 = -\frac{1}{r_{o2}C_{o2}}$$

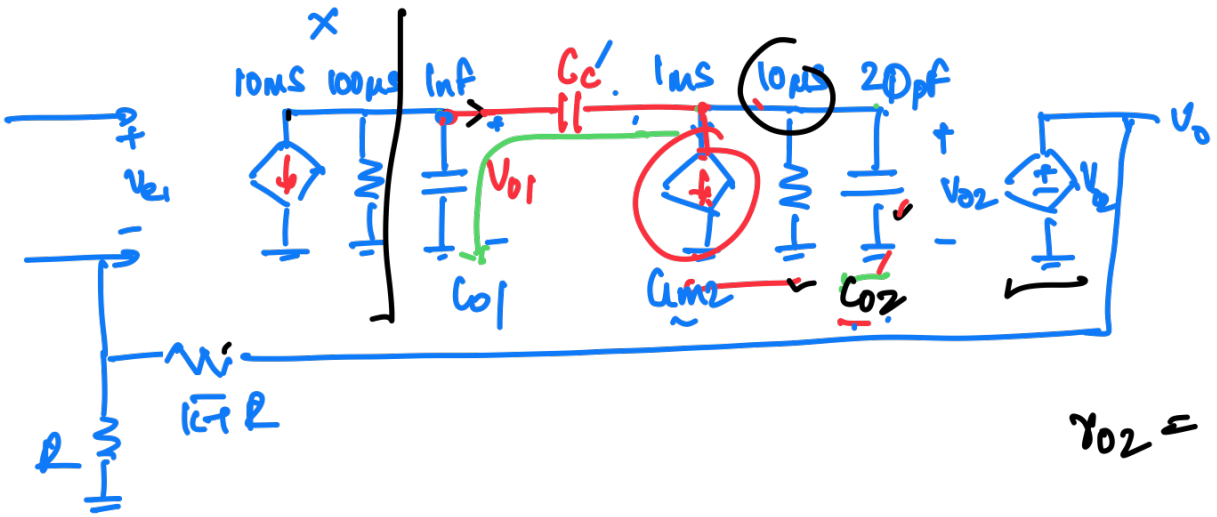
$$p_2' = -\frac{\left((C_{o2}+C_c)\,r_{o2} + \left(C_{o1} + C_c\left(1+g_{m2}r_{o2}\right)\right)r_{o1}\right)}{\left(C_{o1}\left(C_{o2}+C_c\right) + C_cC_{o2}\right)r_{o1}r_{o2}}$$

$$= -\frac{}{\left(C_{o1}+C_c\right)\left(C_{o2} + \frac{C_{o1}C_c}{C_{o1}+C_c}\right)r_{o1}r_{o2}}$$

$\frac{C_{o1}C_c}{C_{o1}+C_c} \approx C_{o1}$

Lecture #17.

$\frac{C_c}{C_c + C_{01}}$



$$r_{o2} = \frac{1}{10\mu s} = 100K$$

$$r_{o1} = \frac{1}{100\mu s} = 10K$$

$$p_1' = \frac{-1}{\left((C_{o2}+C_c)\,r_{o2} + \left(C_{o1} + C_c(1+G_{m2}r_{o2})\right)r_{o1}\right)}$$

$$= \frac{-1}{C_{o1}r_{o1} + C_c(1+G_{m2}r_{o2})\,r_{o1} + (C_{o2}+C_c)\,r_{o2}}$$

$$\approx \frac{-1}{C_c \times G_{m2}\,r_{o2}\,r_{o1}} = p_1''$$

$$p_2' = -\frac{\left((C_{o2}+C_c)\,r_{o2} + \left(C_{o1} + C_c(1+G_{m2}r_{o2})\right)r_{o1}\right)}{(C_{o1}+C_c)\left(C_{o2} + \frac{C_{o1}C_c}{C_{o1}+C_c}\right) r_{o1}\,r_{o2}}$$

$\approx C_{o1}$

$$= \frac{(C_{o1}+C_c)\,r_{o1} + C_c\,G_{m2}\,r_{o2}\,r_{o1} + (C_{o2}+C_c)\,r_{o2}}{\left\{C_{o2} + \frac{C_{o1}C_c}{C_{o1}+C_c}\right\} r_{o1}\,r_{o2}\,(C_{o1}+C_c)}$$

$$= \frac{1 + \frac{C_c}{C_{01}+C_c} g_{m2} \cdot r_{02} + \frac{C_{02}+C_c}{C_{01}+C_c} \frac{r_{02}}{r_{01}}}{\left\{ C_{02} + \frac{C_{01}C_c}{C_{01}+C_c} \right\} r_{02}}$$

$$p_2 = \frac{-1}{r_{02} C_{02}}$$

$$p_2' \approx \frac{\left(\frac{1}{r_{02}} + \frac{C_c}{C_{01}+C_c} g_{m2} \right)}{\left(C_{02} + \frac{C_{01}C_c}{C_{01}+C_c} \right)} \quad ; \; C_c \gg C_{01}, C_{02}$$

$$g_{m2} r_{02} \gg 1$$

$$\frac{r_{02}}{r_{01}} \ll 1$$

$$p_2' \approx \frac{1 + \frac{C_c}{C_{01}+C_c} g_{m2} r_{02} + \frac{C_c}{C_{01}+C_c} \frac{r_{02}}{r_{01}}}{\left(C_{02} + \frac{C_{01}C_c}{C_{01}+C_c} \right) r_{02}}$$

$$\approx \frac{1 + g_{m2} r_{02} + \frac{r_{02}}{r_{01}}}{(C_{02} + C_{01}) r_{02}}$$

$$\approx \frac{g_{m2} r_{02}}{(C_{01}+C_{02}) r_{02}} \approx \frac{g_{m2}}{C_{01}} = p_2''$$

$$A_{CL}(s) = \frac{\frac{1}{K} G_{m1} r_{o1} \; G_{m2} r_{o2} \; \left(1 \mp s C_c / G_{m2}\right)}{\left(1 - s/p_1''\right)\left(1 - s/p_2''\right)}$$

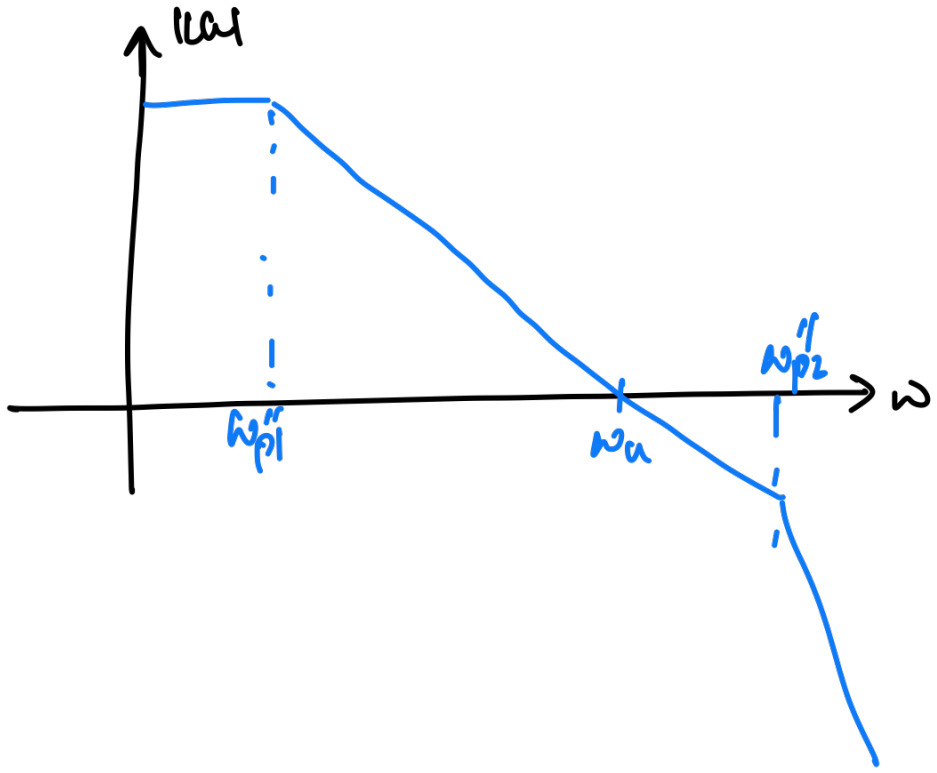
$$\approx \frac{G_{m1} r_{o1} \; G_{m2} r_{o2} / K}{\left[1 + \frac{s C_c (G_{m2} r_{o2}) r_{o1}}{1}\right]\left[1 + s C_{o1} / G_{m2}\right]}$$

Dominant pole, p_1''

Dominant pole freq, $\omega_{p1}'' = \frac{1}{C_c r_{o1} (G_{m2} r_{o2})} = \frac{1}{10 \times 10^{-9} \times 10 \times 10^{3} \times 100} = 10^{2}$

far-away pole freq., $\omega_{p2}'' = \frac{G_{m2}}{C_{o1}} = \frac{1 \times 10^{-3}}{1 \times 10^{-9}} = 10^{6}$

(Non-dominant pole)



$$\omega_u = \frac{(G_{m1} r_{o1} G_{m2} r_{o2})/K}{C_c G_{m2} r_{o2} r_{o1}} = \frac{(G_{m1}/K)}{C_c}$$

$$= \frac{10 \times 10^{-3}}{4 \times 16 \times 10^{-9}}$$

$$= \frac{10^6}{4} = 25 \times 10^4$$

$$\phi_m = \angle L_u - (-180^\circ) = 76^\circ$$

$$= -\tan^{-1}\left(\frac{\omega_u}{\omega_{p1}}\right) - \tan^{-1}\left(\frac{\omega_u}{\omega_{p2}}\right) + 180^\circ = 76^\circ$$

$$-90^\circ + 180^\circ - \tan^{-1}\left(\frac{\omega_u}{\omega_{p2}}\right) = 76^\circ$$

$$\tan^{-1}\left(\frac{G_{m1}/4C_c}{G_{m2}/r_{o1}}\right) = 14^\circ$$

$$\frac{10 \times 25pF}{4 \times C_c} = \tan(14^\circ)$$

$$C_c = \frac{10 \times 25pF}{\tan(14^\circ)} = \underline{10nF}.$$

$10nF \times 100$

$$p_1'' = \frac{-1}{C_{01}r_{01} + C_C(1+G_{m2}r_{02})r_{01} + (C_{02}+C_C)r_{02}}$$

$$= \frac{-1}{C_{01}r_{01}\left\{1 + \frac{C_C}{C_{01}}(1+G_{m2}r_{02}) + \left(\frac{C_{02}}{C_{01}} + \frac{C_C}{C_{01}}\right)\frac{r_{02}}{r_{01}}\right\}}$$

$$= \frac{-1}{C_{01}r_{01}\left\{1 + \frac{C_C}{C_{01}}(101) + \left(\frac{20pF}{1nF} + \frac{C_C}{C_{01}}\right)\frac{100}{10}\right\}}$$

$$= \frac{-1}{C_{01}r_{01}\left\{1 + 101\frac{C_C}{C_{01}} + \left(0.02 + \frac{C_C}{C_{01}}\right)10\right\}}$$

$$= \frac{-\omega_{p1}}{\left\{1.2 + 111\frac{C_C}{C_{01}}\right\}} \quad \checkmark$$

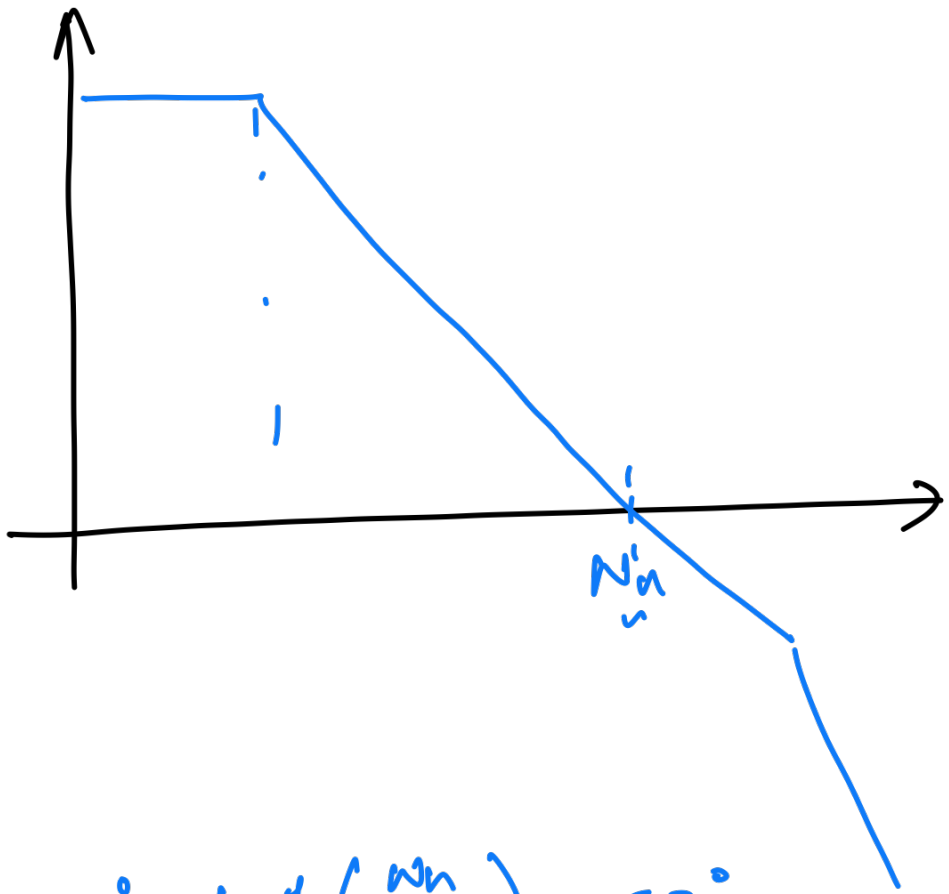
$$p_2'' = -\frac{C_{01}r_{01}\left\{1.2 + 111\frac{C_C}{C_{01}}\right\}}{(C_{01}(C_{02}+C_C) + C_CC_{02})r_{01}r_{02}}$$

$$= -\frac{\left\{1.2 + 111\frac{C_C}{C_{01}}\right\}}{\left\{(C_{02}+C_C) + \frac{C_C}{C_{01}}C_{02}\right\}r_{02}}$$

$$= - \frac{\left\{1.2 + 111\,\frac{C_c}{C_{01}}\right\}}{C_{01}\gamma_{02}\left\{\left(\frac{C_{02}}{C_{01}} + \frac{C_c}{C_{01}}\right) + \frac{C_c}{C_{01}}\,\frac{C_{02}}{C_{01}}\right\}}$$

$$= - \frac{\left\{1.2 + 111\,\frac{C_c}{C_{01}}\right\}}{C_{01}\gamma_{02}\left\{0.02 + \frac{C_c}{C_{01}} + \frac{C_c}{C_{01}} * 0.02\right\}}$$

$$\omega''_{p2} = - \frac{C_{02}}{C_{01}}\,\omega_{p1}\,\frac{\left\{1.2 + 111\,\frac{C_c}{C_{01}}\right\}}{\left\{0.02 + 1.02\,\frac{C_c}{C_{01}}\right\}}$$



$$\phi_m = 90^\circ - \tan^{-1}\left(\frac{\omega_n}{\omega''_{p2}}\right) = 70^\circ$$

$$H_1(s) = \frac{1 - s/\omega_z}{1 + s/\omega_p} \quad ; \quad H_2(s) = \frac{1 + s/\omega_z}{1 + s/\omega_p}$$

$$\omega_p = \frac{\omega_z}{10}$$

$$\mathcal{L}^{-1}\left\{ \frac{1}{s} \frac{1 - s/\omega_z}{1 + s/\omega_p} \right\} = \mathcal{L}^{-1}\left[\frac{\omega_p}{\omega_z} \left\{ \frac{1}{s} + \frac{-\left(\frac{\omega_p + \omega_z}{\omega_p}\right)}{s + \omega_p} \right\} \right]$$

$$\mathcal{L}^{-1}\left\{ \frac{1}{s} \frac{1 + s/\omega_z}{1 + s/\omega_p} \right\} = \mathcal{L}^{-1}\left[\frac{\omega_p}{\omega_z} \left\{ \frac{1}{s} + \frac{-\left(\frac{\omega_z + \omega_p}{\omega_p}\right)}{s + \omega_p} \right\} \right]$$

$$p_1'' = \frac{-w_{p1}}{\left\{1.2 + 111\,\frac{C_c}{C_{o1}}\right\}} = \frac{-10^5}{\left(1.2+111\,\frac{C_c}{C_{o1}}\right)} = -w_{p1}''$$

$$p_2'' = -\frac{C_{o2}}{C_{o1}}\,\frac{w_{p2}\left\{1.2 + 111\,\frac{C_c}{C_{o1}}\right\}}{\left\{0.02 + 1.02\,\frac{C_c}{C_{o1}}\right\}} = -w_{p2}''$$

$$= -\frac{20\times10^{-12}}{10^{-9}} \times 5\times10^5\,\frac{\left(1.2 + 111\,\frac{C_c}{C_{o1}}\right)}{\left(0.02 + 1.02\,\frac{C_c}{C_{o1}}\right)}$$

$$= -\frac{10^5}{10}\,\frac{\left(1.2 + 111\,\frac{C_c}{C_{o1}}\right)}{\left(0.02 + 1.02\,\frac{C_c}{C_{o1}}\right)}$$

$$A_M = \frac{10^4/4}{\left(1+\frac{s}{w_{p1}''}\right)\left(1+s/w_{p2}''\right)}$$

2500×100 rad/s

2.5×10^5

$$w_u \approx \frac{10^4}{4}\,w_{p1}'' = 25\times100 \times \frac{10^5}{\left(1.2+111\,\frac{C_c}{C_{o1}}\right)} = \frac{25\times10^7}{\left(111\left(\frac{C_c}{C_{o1}}\right)+1.2\right)}$$

$$\phi_m = 180^\circ - \tan^{-1}\left(\frac{w_u}{w_{p1}''}\right) - \tan^{-1}\left(\frac{w_u}{w_{p2}''}\right)$$

$$= 180^\circ - \tan^{-1}\left(\frac{10^4}{4}\right) - \tan^{-1}\left(\frac{w_u}{w_{p2}}\right) = 76^\circ$$

$$= 90^\circ - \tan^{-1}\left(\frac{w_u}{w_{p2}''}\right) = 76^\circ$$

$$\frac{\omega_n}{\omega_{p2}''} = \frac{1}{4} = \frac{2500\,\omega_{p1}''}{\omega_{p2}''}$$

$$\frac{1}{4} = 2500 \times \frac{10^4 \Big/ \left(1.2 + 111\frac{C_c}{C_{01}}\right)}{10^4\left(1.2 + 111\frac{C_c}{C_{01}}\right) \Big/ \left(0.02 + 1.02\frac{C_c}{C_{01}}\right)}$$

$$10^5\left(0.02 + 1.02x\right) = \left(1.2 + 111x\right)^2$$

$$= 1.44 + 12321x^2 + 266.4x$$

$$12321x^2 + 266.4x - 102000x - 2000 + 1.44 = 0$$

$$12321x^2 - 101733.6x - 1998.56 = 0$$

$$x = \frac{101733.6 \pm \sqrt{101733.6^2 + 4 \times 12321 \times 1998.56}}{2 \times 12321}$$

$$= \frac{101733.6 \pm 102216.55}{2 \times 12321} = 8.2765$$

$$\frac{C_c}{C_{01}} = 8.2765 \Rightarrow \underline{C_c = 8.2765\,nf}\ \checkmark$$

$$\omega_{p1}'' = \frac{10^5}{\left(1.2 + 111\frac{C_c}{C_{01}}\right)} = \frac{10^5}{1.2 + 111 \times 8.2765} = 108.708\ rad/s$$

$$\omega_{p2}'' = \frac{10^5}{10}\,\frac{\left(1.2 + 111\frac{C_c}{C_{01}}\right)}{\left(0.02 + 1.02\frac{C_c}{C_{01}}\right)} = \frac{10^4\left(1.2 + 111 \times 8.2765\right)}{\left(0.02 + 1.02 \times 8.2765\right)}$$

$$= 10.87081 \times 10^5\ rad/s$$

$$D(s) = s^2\left(C_{01}(C_{02}+C_c)+C_c C_{02}\right)r_{01}r_{02} + s\left((C_{02}+C_c)r_{02} + \left(C_{01}+C_c(1+g_{m2}r_{02})\right)r_{01}\right)+1$$

$$= s^2\left(1000(20+8276.5)+8276.5\times 20\right)10^{-24}\times 100\times10^3\times 10\times10^3 + s\left((20+8276.5)100\times10^3\times10^{-12} + (1000+8276.5\times 111)\times10\times10^3\times10^{-12}\right)+1$$

$$= s^2\left(84620300\times10^{-16}\right) + s\,1002656.5\times10^{-8} + 1$$

$$= \left(\frac{s}{10^8}\right)^2 84620300 + \frac{s}{10^8}\times 1002656.5 + 1 = 0$$

$$\frac{s}{10^8} = \frac{-1002656.5 \pm \sqrt{1002656.5^2 - 4\times 84620300}}{2\times 84620300}$$

$$= \frac{-1002656.5 \pm 1002654.81208}{2\times 84620}$$

$s_1'' = -99.743$ | $\omega_{p1}'' = 99.743$ rad/s ✓

$s_2'' = -11.847\times10^5$ | $\omega_{p2}'' = 11.847\times10^5$ rad/s. ✓

Open loop poles.

	w/o C_c	w/ approx. $C_c = 10nf$	w/ more acc. $C_c = 8.27nf$	w/ $C_c = 8.27nf$ and exact calculation
ω_{p1}	10^5	10^2	108.708 rad/s	99.743 rad/s ✓
ω_{p2}	5×10^5	10^6	10.87×10^5 rad/s	11.847×10^5 rad/s ✓
ϕ_m	3°	76°	76°	

Lecture #18

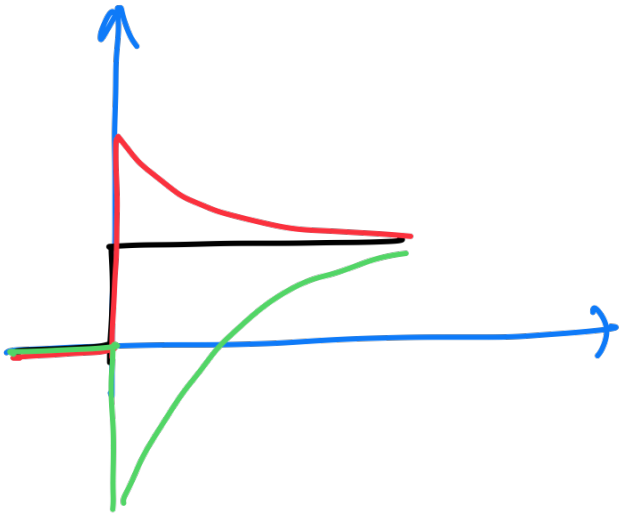
$$H_1(s) = \frac{1+s/\omega_z}{1+s/\omega_p} = \frac{\omega_p}{\omega_z}\,\frac{s+\omega_z}{s+\omega_p}$$

$$H_2(s) = \frac{\omega_p}{\omega_z}\,\frac{\omega_z - s}{\omega_p + s}$$

$$\mathcal{L}^{-1}\left\{\frac{1}{s}H_1(s)\right\} = \mathcal{L}^{-1}\left[\frac{\omega_p}{\omega_z}\,\frac{1}{s}\,\frac{s+\omega_z}{s+\omega_p}\right]$$

$$= \frac{\omega_p}{\omega_z}\,\mathcal{L}^{-1}\left[\frac{\omega_z/\omega_p}{s} + \frac{\frac{\omega_z-\omega_p}{-\omega_p}}{s+\omega_p}\right]$$

$$= \frac{\omega_p}{\omega_z}\left[\frac{\omega_z}{\omega_p}\,u(t) + \left(1-\frac{\omega_z}{\omega_p}\right)e^{-\omega_p t}u(t)\right]$$



$H_1(s)$

$$\frac{\omega_p}{\omega_z}\left[\frac{\omega_z}{\omega_p} + 1 - \frac{\omega_z}{\omega_p}\right]$$

$$\mathcal{L}^{-1}\left\{\frac{1}{s}\,\frac{\omega_p}{\omega_z}\,\frac{\omega_z - s}{\omega_p + s}\right\} = \frac{\omega_p}{\omega_z}\,\mathcal{L}^{-1}\left\{\frac{\frac{\omega_z}{\omega_p}}{s} + \frac{\frac{\omega_z+\omega_p}{-\omega_p}}{s+\omega_p}\right\}$$

$$= \frac{\omega_p}{\omega_z}\left[\frac{\omega_z}{\omega_p}\,u(t) - \left(1+\frac{\omega_z}{\omega_p}\right)e^{-\omega_p t}u(t)\right]$$

at $t=0$, $\quad y(t) = \frac{\omega_p}{\omega_z}\left[\frac{\omega_z}{\omega_p} - \left(1+\frac{\omega_z}{\omega_p}\right)\right]$

$$A_v = \frac{\frac{C_{m1} r_{o1} C_{m2} r_{o2}}{K}\left(1 - \frac{sC_c}{C_{m2}}\right)}{\left(1+\frac{s}{\omega_{p1}''}\right)\left(1+\frac{s}{\omega_{p2}''}\right)}$$

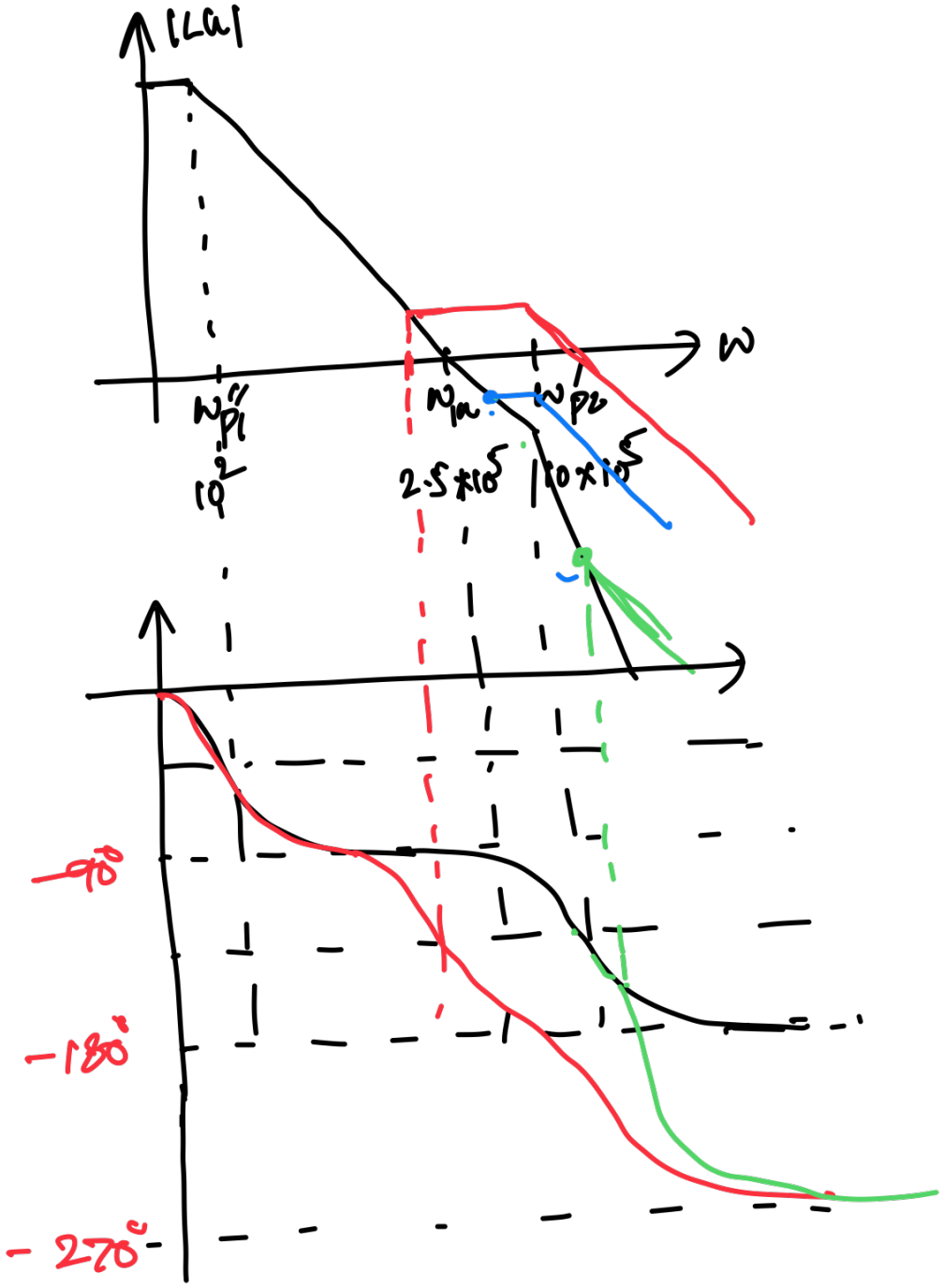
$1 = j\omega\frac{C_c}{C_{m2}}$

$\tan^{-1}\left(\frac{Im}{Re}\right)$

$$\angle A_v = -\tan^{-1}\left(\frac{\omega}{\omega_{p1}''}\right) - \tan^{-1}\left(\frac{\omega}{\omega_{p2}''}\right) - \tan^{-1}\left(\frac{\omega}{\omega_z}\right)$$

$$\phi_m = \angle A_v(\omega_u) - (-180^\circ)$$

$$\omega_z = \frac{C_{m2}}{C_c} = \frac{10^{-3}}{8\times10^{-9}} = 1.25\times10^5 \text{ rad/s.}$$



$a+jb$ $\quad (a,b>0)$

$\tan^{-1}\left(\frac{b}{a}\right)$

$a-jb$

$-\tan^{-1}\left(\frac{b}{a}\right)$

$$N(s) = g_{m1} r_{o1} g_{m2} r_{o2} \left(1 - \frac{sC_c}{g_{m2}}\right)$$

$$= g_{m1} r_{o1} g_{m2} r_{o2} \left(1 - \frac{1}{g_{m2}} \times \frac{1}{\frac{1}{sC_c} + R_x}\right)$$

$$= g_{m1} r_{o1} g_{m2} r_{o2} \left(1 - \frac{1}{g_{m2}} \frac{sC_c}{1 + sC_c R_x}\right)$$

$$\Downarrow$$

$$\sim \frac{\left\{g_{m2}(1 + sC_c R_x) - sC_c\right\}}{1 + sC_c R_x}$$

$$\left\{\frac{g_{m2} + sC_c(g_{m2}R_x - 1)}{g_{m2}(1 + sC_c R_x)}\right\} \quad = 0$$

$$g_{m2} = \frac{1}{R_x}$$

$$N(s) = g_{m1} r_{o1} g_{m2} r_{o2} \times \frac{1}{(1 + sC_c R_x)}$$

if $g_{m2} R_x > 1 \Rightarrow \omega_z = \frac{g_{m2}}{(g_{m2} R_x - 1) C_c} \Rightarrow g_{m2} R_x > 1$

$$R_x > \frac{1}{g_{m2}}$$

$$g_{m2} > \frac{1}{R_x}$$

$$\frac{g_{m2}}{C_c} > \frac{1}{R_x C_c}$$

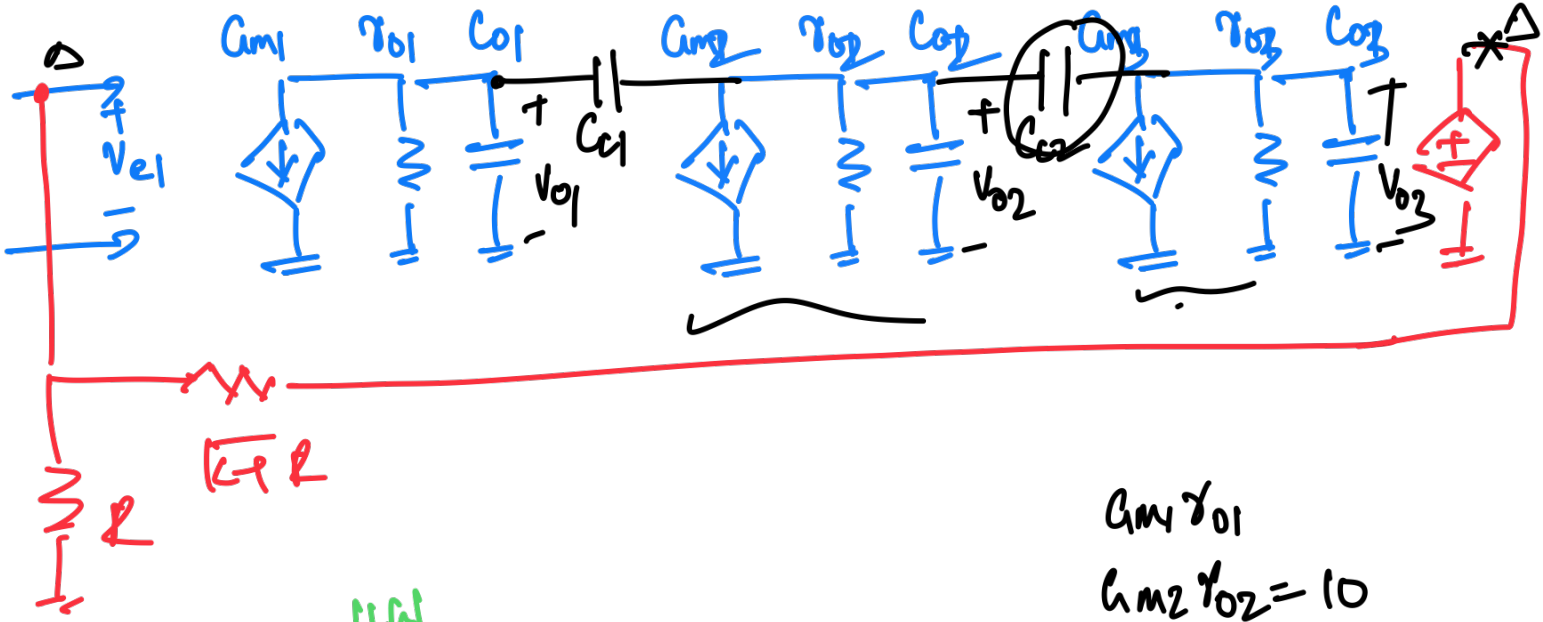
$$D(s) = s^2\left(C_{01}(C_{02}+C_c)+C_cC_{02}\right)r_{01}r_{02}$$
$$+s\left((C_{02}+C_c)r_{02}+\left(C_{01}+C_c(1+g_{m2}r_{02})\right)r_{01}\right)+1$$

$$= \left\{sC_{01}(sC_{02}+sC_c)+sC_{02}\cdot sC_c\right\}r_{01}r_{02}$$
$$+\left(sC_{02}+sC_c\right)r_{02}+sC_{01}r_{01}+sC_c\left(1+g_{m2}r_{02}\right)r_{01}+1$$

$$= \left\{sC_{01}\cdot\left(sC_{02}+\frac{sC_c}{1+sC_cR_x}\right)+sC_{02}\cdot\frac{sC_c}{1+sC_cR_x}\right\}r_{01}r_{02}$$
$$+\left(sC_{02}+\frac{sC_c}{1+sC_cR_x}\right)r_{02}+sC_{01}r_{01}+\frac{sC_c}{1+sC_cR_x}\left(1+g_{m2}r_{02}\right)r_{01}+1$$

$$\frac{1}{sC_c} = R_x+\frac{1}{sC_c} = \frac{1+sC_cR_x}{sC_c}$$

$$\left(1+\frac{s}{p_1}\right)\left(1+\frac{s}{p_2}\right)\left(1+\frac{s}{p_3}\right)$$



$G_{m1} r_{o1}$

$G_{m2} r_{o2} = 10$

$G_{m3} r_{o3} = 50$

